Cartan's Magic Formula

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Abstract

Short notes on Cartan's magic formula

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1 Lie Derivatives

Given a smooth map $F : M \to N$ between manifolds and a point $p \in M$ with image $q = F(p) \in N$, we define the pushforward map

$$F_* : T_pM \to T_qN \tag{1}$$

as follows. For $v_p \in T_pM$, take any curve $\gamma : (-1, 1) \to M$ in the equivalence class of $v \in T_pM$

$$\gamma(0) = p, \quad \dot{\gamma}(0) = v_p \tag{2}$$

and consider the image curve $F \circ \gamma : (-1, 1) \to N$, which has $(F \circ \gamma)(0) = q$ by construction. The pushforward $F_*(v_p)$ is defined to be the equivalence class of this image curve. We define the more general bundle map $F_* : TM \to TN$ to act pointwise in this way. By taking tensor products at each point and applying linearity, this definition extends to higher powers of the respective tangent bundles.

Under the same assumptions we define the pullback

$$F^* : T^*_p(N) \to T^*_p(M) \tag{3}$$

by the rule

$$F^*(\omega_q)(v_p) = \omega_q(F_*(v_p)), \quad \omega_q \in T^*_q(N). \tag{4}$$

In the same way as the pushforward, the pullback at any one cotangent space to N naturally extends to a bundle map $F^* : T^*N \to T^*M$. And as before, this extends further to higher powers of the cotangent bundles.

In general there is no well-defined and unique way to pullback a vector, essentially because a given vector in T_qN may not be the image of any vector in T_pM; analogous obstructions apply to the pushforward a 1-form. However, if F is a diffeomorphism (that is, a bijection which is smooth in both directions), then we can define the pullback of a vector under F to be its pushforward under F^{-1}, and similarly for 1-forms. Taking products, this lets us push and pull tensors of arbitrary ranks in either direction.

Next, given any vector field X on a smooth manifold M, we define the local flow

$$\varphi^t_X : M \to M, \quad \varphi^0_X(p) = p, \frac{d}{dt}\varphi^t_X(p) = X(p). \tag{5}$$

Given a rank $(0, s)$ tensor field T (that is, one built up from tensor products of the cotangent bundle), we define the Lie derivative as the linear response to pushing forward under this flow. Concretely, at a point $p \in M$ we have

$$(\mathcal{L}_X T)_p := \lim_{\varepsilon \to 0} \frac{\left((\varphi_X^\varepsilon)^* T \right)_p - T_p}{\varepsilon} = \lim_{\varepsilon \to 0} \frac{(\varphi_X^\varepsilon)^* T_{\varphi_X^\varepsilon p} - T_p}{\varepsilon}. \tag{6}$$

Even more concretely, this is a multilinear map on $T_p M$ which acts on vectors $v_p^{(1)}, \ldots, v_p^{(s)} \in T_p M$ as

$$\begin{aligned} (\mathcal{L}_X T)_p \left(v_p^{(1)}, \ldots, v_p^{(s)} \right) &= \lim_{\varepsilon \to 0} \frac{(\varphi_X^\varepsilon)^* T_{\varphi_X^\varepsilon p} \left(v_p^{(1)}, \ldots, v_p^{(s)} \right) - T_p \left(v_p^{(1)}, \ldots, v_p^{(s)} \right)}{\varepsilon} \\ &= \lim_{\varepsilon \to 0} \frac{T_{\varphi_X^\varepsilon (p)} \left((\varphi_X^\varepsilon)_* \left(v_p^{(1)} \right) \cdots (\varphi_X^\varepsilon)_* \left(v_p^{(s)} \right) \right) - T_p \left(v_p^{(1)}, \ldots, v_p^{(s)} \right)}{\varepsilon}. \end{aligned} \tag{7}$$

2 p-forms in Coordinates

Consider in particular $\omega \in T^* M$ and $p \in M$. Then we have

$$(\mathcal{L}_X \omega)_p (v_p) = \lim_{\varepsilon \to 0} \frac{\omega_{\varphi_X^\varepsilon (p)} \left((\varphi_X^\varepsilon)_* (v_p) \right) - \omega_p (v_p)}{\varepsilon}. \tag{8}$$

In local coordinates x^μ, the nearly-identity diffeomorphism φ_X^ε acts as

$$p \mapsto p', \quad x^\mu (p') = x^\mu (p) + \varepsilon X^\mu (p) \tag{9}$$

so our the innermost object appearing in our expression for $(\mathcal{L}_X \omega)_p (v_p)$ has components

$$(\varphi_X^\varepsilon)_* (v_p) \in T_{p'}, \quad ((\varphi_X^\varepsilon)_* (v_p))^\mu = (v_p)^\mu + \varepsilon (\partial_\nu X^\mu)_p v_p^\nu + o\left(\varepsilon^2\right) \tag{10}$$

while the outermost object has components

$$\omega_{\varphi_X^\varepsilon (p)} \equiv \omega_{p'}, \quad (\omega_{p'})_\mu = (\omega_p)_\mu + \varepsilon (\partial_\nu \omega_\mu)_p X_p^\nu + o\left(\varepsilon^2\right). \tag{11}$$

Assuming enough analyticity to do the natural expansion in powers of ε, we therefore find

$$\begin{aligned} (\mathcal{L}_X \omega)_p (v_p) &= \lim_{\varepsilon \to 0} \frac{\left[(\omega_p)_\mu + \varepsilon (\partial_\nu \omega_\mu)_p X_p^\nu \right] \left[(v_p)^\mu + \varepsilon (\partial_\nu X^\mu)_p v_p^\nu \right] - \omega_p (v_p)}{\varepsilon} \\ &= (\partial_\nu \omega_\mu)_p X_p^\nu v_p^\mu + (\omega_p)_\mu (\partial_\nu X^\mu)_p v_p^\nu \\ &= (\partial_\nu \omega_\mu)_p X_p^\nu v_p^\mu + (\omega_p)_\nu (\partial_\mu X^\nu)_p v_p^\mu. \end{aligned} \tag{12}$$

As this holds at an abstract point and with a general input, we have more generally

$$(\mathcal{L}_X \omega)_\mu = X^\nu \partial_\nu \omega_\mu + \omega_\nu \partial_\mu X^\nu. \tag{13}$$

An identical calculation for ω a 2-form gives

$$(\mathcal{L}_X \omega)_{\mu\nu} = X^\rho \partial_\rho \omega_{\mu\nu} + \omega_{\rho\nu} \partial_\mu X^\rho + \omega_{\mu\rho} \partial_\nu X^\rho. \tag{14}$$

On a 3-form we find

$$(\mathcal{L}_X \omega)_{\mu_1 \mu_2 \mu_3} = X^\rho \partial_\rho X_{\mu_1 \mu_2 \mu_3} + \omega_{\rho \mu_2 \mu_3} \partial_{\mu_1} X^\rho + \omega_{\mu_1 \rho \mu_3} \partial_{\mu_2} X^\rho + \omega_{\mu_1 \mu_2 \rho} \partial_{\mu_3} X^\rho \tag{15}$$

and so on.

3 The Magic Formula

Given a p-form ω and a vector v, we denote by $\omega \cdot v$ the $(p-1)$-form given by placing v in the first slot of ω. That is,

$$(v \cdot \omega)(A_1, \ldots, A_{p-1}) := \omega(v, A_1, \ldots, A_{p-1}). \tag{16}$$

This is sometimes written $\iota_v(\omega)$, and sometimes called the interior product. As for a general tensor, recall that the components of a p-form in some local coordinate system are by definition the numbers they return when acting on tuples of basis vectors.

$$\omega_{\mu_1 \cdots \mu_p} := \omega(\partial_1, \ldots, \partial_p). \tag{17}$$

To refer to basis p-forms, we write $[\cdot]$ for the averaging antisymmetrizing bracket which leaves already-antisymmetric object unchanged, *e.g.*,

$$A_{[\mu\nu]} = \frac{A_{\mu\nu} - A_{\nu\mu}}{2!}. \tag{18}$$

The wedge product of forms then has components

$$(\omega \wedge \sigma)_{\mu_1 \cdots \mu_p \nu_1 \cdots \nu_q} := \frac{(p+q)!}{p!q!} \omega_{[\mu_1 \cdots \mu_p} \sigma_{\nu_1 \cdots \nu_q]} \tag{19}$$

and the exterior derivative of a single form has components

$$(d\omega)_{\mu_0 \cdots \mu_p} = (p+1) \partial_{[\mu_0} \omega_{\mu_1 \cdots \mu_p]}. \tag{20}$$

For example, at $p = 1$ we have

$$(d\omega)_{\mu\nu} = \partial_\mu \omega_\nu - \partial_\nu \omega_\mu. \tag{21}$$

With some inspired massaging, we can therefore rearrange our expression for the Lie derivative of a 1-form as

$$\begin{aligned}
(\mathcal{L}_X \omega)_\mu &= X^\nu \partial_\nu \omega_\mu + \omega_\nu \partial_\mu X^\nu \\
\implies (\mathcal{L}_X \omega)(A) &= X^\nu A^\mu \partial_\nu \omega_\mu + \omega_\nu A^\mu \partial_\mu X^\nu \\
&= X^\nu A^\mu \partial_\nu \omega_\mu - X^\nu A^\mu \partial_\mu \omega_\nu + A^\mu \partial_\mu (\omega_\nu X^\nu) \\
&= (d\omega)(X, A) + d(X \cdot \omega)(A) \\
&= (X \cdot d\omega + d(X \cdot \omega))(A)
\end{aligned} \tag{22}$$

or without the test function

$$\mathcal{L}_X \omega = X \cdot d\omega + d(X \cdot \omega). \tag{23}$$

This is called Cartan's magic formula. Similar checks show that it holds for general p-forms.